Constraint-first string theory: the flux landscape's only exact structure is its constraint

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Abstract

We measure, on real supersymmetric flux vacua, which low-energy observables carry information about the flux that produced them. Every flux-determined observable is classified by a joint (multi-coordinate quadratic) learnability test, gated by a positive control that must reproduce the one exactly-known quantity. The result is a *structure map*: the string coupling, the gauge couplings, the mass-spectrum *shape*, the axions, the dilaton, and the cosmological constant are all statistically uniform in flux ($r < 0.25$); W_0 and the saxions carry weak structure ($r \approx 0.3$–0.6); and exactly one quantity is *exactly* structured — the D3 tadpole, the symplectic pairing $N_{\text{flux}} = F^\top J H$ ($r = 1.0$). No observable carries the strong ($r > 0.9$) *geometric fingerprint* that would pin the compactification. We conclude: the landscape's only exact structure is what it *forbids*, not what it *selects*; everything it produces is noise plus a weak scale. This is the constraint-first program, established by measurement rather than asserted.

1 Introduction

The string landscape is $\sim 10^{500}$ vacua and no experiment that tells most of them apart. Two programs have tried to make it navigable. The *search* program (and its modern machine-learning descendants) asks: can we *learn* the map from flux to observables, and so scan our way to our universe? The *constraint* program (swampland, bootstrap) asks instead: which consistency conditions *eliminate* vacua? This paper measures the premise that distinguishes them: whether any observable carries a *fingerprint* — strong information about the flux — that a search could exploit.

The answer, measured on 174 supersymmetric IIB flux vacua, is that no observable does. The only exact structure is the constraint itself.

2 Method

Sample. JAXVacua [1] on a bundled $h^{1,2} = 2$ flux EFT, the natural ISD measure (`isd_mode` = `ISD−`, Gaussian flux prior), degree-0 spectrum. 174 supersymmetric vacua (selected by $DW = 0$, $\mathrm{Im}\,\tau > 0$) from 1318 critical points.

The joint test and the Rule of Three. For each observable y we compute the cross-validated ridge regression of y on the degree-2 monomial basis of the 12 flux coordinates (91 features), reporting the Pearson r between held-out predictions and truth. A single-coordinate mutual-information test is *blind* to multi-coordinate structure and is used only as a first screen; the claim follows the *Rule of Three*: single-coordinate MI, joint ridge-r, and a permutation null must *agree*. A disagreement is unresolved, never resolved by one test.

The control. The D3 tadpole is *known* to be the symplectic pairing $N_{\text{flux}} = F^\top J H$ with $J = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, a clean bilinear form of the flux. The joint test must return $r = 1.0$ for it, at machine precision. This control is what makes every other reading trustworthy: a test that cannot see the tadpole's structure cannot be trusted to see any.

3 Results: the structure map

observable	wide-flux r	verdict
tadpole N_{flux}	**1.00**	**exact** (the constraint)
W_0	0.62	weakly learnable (scale via tadpole)
saxions (moduli sizes)	0.30–0.42	weakly learnable (stabilization)
string coupling g_s	0.12	uniform
gauge couplings	0.02–0.12	uniform
mass-spectrum shape (ratios)	<0.25	uniform
axions (moduli phases)	0.15	uniform
axio-dilaton $\mathrm{Re}\,\tau$	0.13	uniform
AdS potential V	0.00	uniform

The pattern is stark. The low-energy observables are a mostly-unstructured cloud: the string coupling, the gauge couplings, the mass-spectrum *shape*, the axions, the dilaton, and the cosmological constant are statistically uniform in flux. Two things carry weak structure — W_0, whose *scale* tracks the tadpole, and the saxions, which flux stabilizes. Exactly one thing is *exactly* structured: the tadpole, the constraint.

Remark 1. ***No geometric fingerprint exists.*** *Across every observable measured, none carries strong ($r > 0.9$) information about the flux. The fingerprint that a search program would need — an observable that pins the compactification — is absent. The only exact structure is what the landscape* forbids*.*

4 The W_0 correction and the search

A first pass reported W_0 as "zero-information," on the strength of a single-coordinate correlation; that is retracted here. The joint test gives $r = 0.62$ on the natural measure ($r = 0.10$ only when flux is artificially bounded). W_0 is weakly learnable, through its scale.

This does not rescue search. On the sample, $\log|W_0|$ has standard deviation 0.63 nats, so a model with $r = 0.62$ has prediction error $\sigma_{\text{pred}} = 0.50$ nats. The KKLT tuning target $\log|W_0| = -9.2$ sits 9.9 nats — about $20\,\sigma_{\text{pred}}$ — below the mean. A model fit on the data cannot guide a search to a point twenty prediction-errors away; that is a meaningless extrapolation, not a gradient.

Remark 2. ***Search over W_0 is hopeless by rarity plus extrapolation****, not by zero-information. W_0 is log-uniform (CDF $R^2 = 0.87$–0.92), so small W_0 is exponentially rare; and the target is $20\,\sigma_{\text{pred}}$ outside the observed range. The conclusion survives; the argument is now measured.*

5 The swampland connection

The constraint-first claim is the swampland program stated as a measurement. Every swampland conjecture is a "consistency *forbids* X" statement — the interesting structure lives in the constraints,

not in what the theory produces. The structure map makes the same claim empirically, so each conjecture becomes a falsifiable prediction on the same samples (`pacland.swampland`, [5]):

conjecture	quantity	ridge r	verdict
Weak Gravity (electric)	$g_{\min}$ = lightest coupling	-0.13	uniform
Weak Gravity (electric)	g_{mean}	-0.16	uniform
No global symmetries	charge gcd	–	consistent ($= 1$)
Distance (proxy)	$\log m_{\text{light}}$ vs distance	-0.68	real signal
Gravity/tadpole (anchor)	N_{flux}	$+0.98$	structured

The Weak-Gravity scale $\Lambda = g\,M_{\text{pl}}$ depends on g_s through the stabilized moduli, so the conjecture predicts it is uniform in flux — and it is ($r < 0.25$, Rule of Three). The charge lattice shows no common divisor (gcd = 1): no forced discrete global symmetry. The distance-conjecture proxy is a *real signal*, reported with its bounds: the lightest mass falls with the complex-structure moduli distance from the sample centroid (Pearson $r = -0.68$, $z = -9.2$; rank-Spearman $r = -0.46$, $z = -6.3$), and it survives flux control (residual $r = -0.63$ after removing the ridge-quadratic flux prediction). It is directionally what the distance conjecture predicts, but it runs on the complex-structure moduli — where the conjecture does not apply — with a centroid-Euclidean distance, so the clean test is queued behind the full Kähler sector. The tadpole anchor ($r = 0.98$, ridge $z = 9.1$) is the positive control; note its single-coordinate MI is only $z = 1.2$ — MI is blind to a bilinear form, which is exactly why the Rule of Three requires the joint test.

These are *predictions*, not new deficit rungs: the swampland bounds have no experimental target value, so they extend the falsifiable surface, and the verification deficit remains $m = 5$.

6 Conclusion

The constraint-first program is established by measurement:

¿ **The flux landscape's only exact structure is its constraint.** Everything ¿ the landscape *produces* — couplings, masses, phases, the cosmological ¿ constant — is noise plus a weak scale; the one thing that is exactly determined ¿ is the D3 tadpole, the quantity that *rules vacua out*. No search can find our ¿ vacuum in observables, because there is nothing structured to find; the only ¿ navigable move is elimination by verified constraint.

Scope, named. One model ($h^{1,2} = 2$), degree-0 spectrum, 174 vacua, flux coordinates only, the IIB flux corner. The instanton-corrected (Kähler-moduli) masses remain computationally infeasible. Each of these is a named limit, not a hidden one; the structure map is a measurement to be reproduced, and the Rule of Three is the instrument that keeps it honest.

References

[1] A. Dubey, S. Krippendorf, A. Schachner, "JAXVacua" (arXiv:2306.06160).

[2] F. Denef and M. R. Douglas, "Distributions of flux vacua," JHEP **0405** (2004) 072.

[3] L. G. Valiant, "A theory of the learnable," CACM **27** (1984) 1134.

[4] L. G. Valiant, "Evolvability," J. ACM **56** (2009) 3.

[5] E. Palti, "The Swampland: Introduction and Review," Fortsch. Phys. **67** (2019) 1900037.